STAFF REVIEW · H446 310 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Saved data for an active function call.

2. On return from the deeper call.

3. O(n) call stack.

4. 3→5; index5 found.

I Do — source worked steps

1. Explain the result — Outputs 321 on descent. Nothing prints for n0.

We Do — guided CHECK

1. Explain the result
Outputs 123 on return as calls unwind.

You Do — source mark schemes

1. Calls 4,3,2,1,0 [2], contributions 0,3,0,1 [2], base0 [1], result4 [1].

2. Print 4 [1],3 [1],2 [1],1 [1] on descent [1]; base0 prints nothing [1].

3. Initialise sum=0 [1]; iterate odd values using for(int i=1;i<=n;i+=2) [2]; add each i to sum and return it [1]; both approaches take O(n) time [1]; iterative O(1) auxiliary memory versus O(n) recursive call stack [1].

Exit Ticket — source answer

1. True. The deeper call returns before printing.